

AGROCLIMATOLOGY

Water Transport Characteristics of Cotton as Affected by Drip Irrigation Layout

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ABSTRACT

To reduce drip irrigation equipment costs, dripper lines are laid in alternate interrow spaces. This study examined the effect of this practice on cotton (*Gossypium hirsutum* L.) water relations. Measurements of water uptake and water potential of plants grown in the field were conducted during four rainless summers at two locations. During the first summer, drippers in every interrow space delivered about 50 mm of water, the exact amount depending on estimated evapotranspiration (ET), every 10 d. The following summer, drip lines were laid in alternate interrow spaces, applying frequent (2–4 d) and small (10–20 mm) amounts of water, based on estimated ET. Treatments were repeated the following two summers. Measurements included water uptake (F), leaf water potential (ψ_l), and soil water content. Whenever a linear relationship between diurnal ψ_l and F was found, we determined the hydraulic conductance (K_T) of the plants as the line's slope reciprocal, and the intercept (ψ_0) as the water potential at the root–soil interface. A linear relation was found in the treatment with drippers in every interrow space. Water stress developed toward the end of the irrigation interval, causing K_T to decrease without affecting ψ_0 . Irrigation restored K_T to its initial value. With the alternate interrow-space pattern, the relationship between ψ_l and F remained linear only until midday. Afternoon (ψ_l , F) points were below those of the morning, suggesting a lowering of the soil water potential near the roots. It was concluded that drippers laid in alternate interrow spaces expose cotton to frequent short periods of water stress.

A DRIP IRRIGATION SYSTEM offers easy control and dosage of water application (Dasberg and Bresler, 1985). In row field crops, laying one lateral dripper line in alternate interrow spaces reduces cost. Although this practice is commonly used in Israel, its effect on cotton yield has not been documented. Such dripper spacing confines the wetted soil volume to a small fraction of the ground surface. Frequent irrigation is assumed to compensate for the smallness of the water reservoir in the soil and to maintain a high soil water potential near the roots, thus preventing water stress (Dasberg and Bresler, 1985).

Plant-water status depends on the balance between the influx from the soil and the efflux to the atmosphere. Water moves through the plant conducting system in proportion to the driving force, $\Delta\psi$, a water potential difference (MPa), and the hydraulic conductance (K_T , $\text{mm h}^{-1} \text{MPa}^{-1}$). For steady-state conditions, the water

uptake F (mm h^{-1}) is simply

$$F = K_T (\psi_0 - \psi_l) \quad [1]$$

where ψ_0 is the average water potential at sites of entry into the roots, possibly at the soil–root interface, and ψ_l is the average exit potential in the leaves. Under laboratory conditions, it is possible to establish steady-state flow to measure K_T and to study the effect of water supply in the soil on the magnitude of K_T . Using this experimental approach with fritted clay as a root medium, Lascano and Van Bavel (1984) found that K_T for a given cotton plant was independent of the leaf water potential and the transpiration rate. They did not verify experimentally the effect of root medium water deficit on this conductance; however, the constant value served to predict water uptake of a plant with roots split between wet and dry compartments, with the contribution of the two compartments weighted by their respective water potential and root length.

Applying the steady-state approximation in the field, where atmospheric evaporative conditions undergo a diurnal cycle, Jones et al. (1982) and Hirasawa and Ishihara (1991) reported that the hydraulic conductance of plants increased with increasing transpirational flux. In their analysis, they assumed that ψ_0 was the bulk soil water potential. Rieger and Motisi (1990) tested the hypothesis that ψ_0 is the bulk soil water potential under steady-state conditions, and also found increasing conductances at higher rates of transpiration. Relative to the water potential measured in leaves, the bulk soil water potential in wet soil is near zero, transforming Eq. [1] into

$$K_T = -F/\psi_l \quad [2]$$

For the situations where $F = 0$, Eq. [1] shows that $\psi_0 = \psi_l$. Accordingly, laboratory measurements by Lascano and Van Bavel (1984), Passioura and Munns (1984), Baker and Van Bavel (1988), and Rieger and Motisi (1990) under steady-state conditions, and by Cohen et al. (1983) and Moreshet et al. (1990) in the field, set the value of ψ_0 around -0.5 MPa. Therefore, omitting ψ_0 in Eq. [2] underestimates K_T , especially at high values of the leaf water potential, invalidating the conclusions of Jones et al. (1982) and Hirasawa and Ishihara (1991).

To account for these results, the electrical network analog more appropriate than the linear battery–resistor model is a diode with ψ_0 as the threshold potential and with conductance defined as

$$K_T = -dF/d\psi_l \quad [3]$$

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In support of this model, measurements of Hirasawa and Ishihara (1991) showed that K_T increases and approaches a constant value as F increases.

To adjust to the fluctuation of the evaporative drive, leaf water potential and water uptake change during the day. Since plant tissues have finite hydraulic capacitance, defined as the rate of change of relative water content (dRWC) per unit of leaf water potential change ($\Delta\psi_l$), water depletion and recharge in the plant may become a significant term not accounted for in the formulation of Eq. [1]. The lag of the diurnal evolution of ψ_l behind transpiration (Reicosky et al., 1982; Sharratt et al., 1983; Katerji et al., 1986) provides evidence for the presence of a capacitor in the electrical analog network.

Measuring water uptake at the base of the stem may lessen the role of water storage organs along the water flow pathway through plants. With sap flow measured at the base of the stem near the water uptake sites in the soil, Berger (1971) observed occasionally that the relation between diurnal flow and leaf water potential was linear; however, he also showed examples of typical diurnal loops on warm days. Since Berger (1971) used a self-made heat-pulse technique that did not compensate for the 15°C diurnal temperature fluctuation of soil water and stem on warm days, he overestimated uptake during the morning warming period and underestimated it during the afternoon. Cohen et al. (1983), using a temperature drift-compensated heat-pulse method, found that diurnal stem sap flow and leaf water potential in citrus trees (*Citrus* spp.) are tightly linear, suggesting that the hydraulic capacitance of the root system is low. Thus, measuring the water uptake by a sap flow tracer method at the base of the stem reduces the effect of plant tissue hydraulic capacitance on transport and allows using Eq. [1] to relate diurnal flow and potential. Still, the steady-state assumption also implies that water supply potential, ψ_0 , remains constant during a day. It should be noted that cotton leaves may dehydrate during the day, affecting ψ_l , which means that the assumption of steady state may not apply to all cases.

The rationale behind frequent irrigation is to maintain a nearly constant high water potential in the root zone; however, widely spaced drippers may concentrate roots in small soil volumes. Water depletion from the small wet volume may be fast enough to change soil water potential and soil hydraulic conductivity near the soil-root interface and decrease ψ_0 or K_T , or both. Since these parameters regulate the water transport to the evaporation sites in the leaves, the decrease will induce leaf water stress, which will appear as a decrease of stomatal conductance and lower transpiration rate.

Our objectives were to examine whether the practice of wetting only a small volume of the soil, as happens with sparse spacing of dripper lines, induces water stress in cotton plants, affecting their water balance and, if so, to detect the causes of the water stress.

MATERIALS AND METHODS

Experimental Fields

The experiments were conducted from 1990 to 1993 in several commercial and experimental cotton (*Gossypium hirsu-*

tum L. cv. SJ 2) fields in Israel, at Na'an (31°52' N, 34°51' E; 85 m above mean sea level) and Nahal Oz (31°29' N, 34°30' E; 80 m above mean sea level). Plant density varied between 90 000 to 100 000 ha⁻¹, with an in-row spacing of 0.96 m. The soil at Na'an was a loamy clay (41% clay, 46% sand), type Typic Haploxerert, and the soil in Nahal Oz was a loamy soil (22% clay, 53% sand), type Calcic Haploxeralf.

Irrigation was by surface drippers placed 1 m apart in-line. Lines were laid in every interrow space for the high-density treatments or in alternate interrow spaces for the low-density treatments. The experimental treatments are summarized in Table 1.

Measurements

Climate

A meteorological station in the middle of the experimental fields measured global irradiance (CM-11, Kipp & Zonen, Delft, Netherlands), air temperature and relative humidity (Skye Instruments 1012, Llandrindod Wells, UK, and Campbell Scientific, Logan, UT, USA, Model 207), and wind speed (Campbell Scientific, Model 014A), all at 2 m above ground level. Potential transpiration was computed using the procedure given by Fuchs et al. (1987).

Soil Moisture

A calibrated neutron scattering probe (Ronli Electronics, Rishon LeZion, Israel) measured soil water content every 0.3 m to a depth of 1.5 m. In 1990, three tubes per row in each of 10 rows were installed. Measurement frequency was 3 to 4 d. In 1993, five tubes, one per row, were installed in each of the two treatments, and measured on each of the days of intensive physiological data collecting. Estimates of soil water potentials were derived from retention curves measured in the laboratory using pressure plates. Direct soil water potential was measured in 1992 at two stations per plot using tensiometers that were embedded down to 0.9 m at 0.3-m intervals. No soil moisture measurements were taken in 1991.

Water Uptake

The heat-pulse method (Cohen et al., 1988) was used to measure sap flow. Sensors were installed at the bottom of the stem. Average stem diameter was estimated from measurements of 50 plants randomly chosen. Sensors were installed on plants with stem diameter within 5% of the average. Ten plants were used in the 1990 experiment. We measured the sap flow of five plants in each of two plots in 1991 and 1992. Ten sensors per plot were used in 1993. Plant population density was used to convert hourly water uptake per plant to millimeters per hour.

Leaf Water Potential

A pressure chamber (Arimad-2, Kefar Haruv, Israel) was used at least hourly on five sunlit leaves and five shaded leaves sampled from five plants near those measured by heat pulse. If a measurement deviated by more than 0.1 MPa from the average, one or two additional leaves were included in the sample. Time of measurements was recorded and the potentials were averaged to correspond to the time of water uptake measurements.

ψ_l Computations

Both the entrance of water into the plant and its exit from the plant are distributed in space. The definition of the water

Table 1. Irrigation treatments in a study of water transport characteristics of cotton as affected by drip irrigation layout, at two locations in Israel.

Location	Year	Dripper line layout in interrows	Dripper outflow	Irrigation interval	Irrigation amount
			L h ⁻¹	d	mm
Na'an	1990	every	2.3	10	50
Nahal Oz	1991	alternate	2.3	3–4†	10–20‡§¶
Nahal Oz	1992	every	2.3	10–14†	50–80‡
Nahal Oz	1992	alternate	2.3	10–14†	50–80‡
Nahal Oz	1993	every	1.2	2–4†	10–20‡#
Nahal Oz	1993	alternate	2.3	2–4†	10–20‡#

† Depending on water availability.

‡ Adjusted to the estimated evapotranspiration based on Class A pan evaporation.

§ Irrigation withheld from Day of Year (DOY) 197 until DOY 205.

¶ Irrigation withheld from DOY 205 until DOY 218. Water applied on DOY 205 did not replenish the previous depletion.

Irrigation withheld from DOY 213 until DOY 224.

potential at both ends is therefore statistical, not geometrical. Thus, the ψ_l was computed as the leaf area-weighted average of sunlit and shaded leaf water potentials (Moreshet et al., 1990), with the assumption that both are transpiring (Petersen et al., 1991):

$$\psi_l = \frac{\psi_{\text{sunlit}} L^*}{L} + \frac{\psi_{\text{shade}} (1 - L^*)}{L} \quad [4]$$

The sunlit leaf area index, L^* , is obtained from measured L (leaf area index) and the zenith angle of the sun, h , for a uniform canopy (Lemur, 1973):

$$L^* = [1 - \exp(-kL)]/k \quad [5]$$

where k is the extinction coefficient. For a spherical leaf distribution,

$$k = 1/(2 \cos h) \quad [6]$$

The leaf area index L was estimated from leaf area measurements on plants from 1-m lengths of row; L was never less than 3, with canopy fully covering the ground.

RESULTS

Measurements of water uptake, F , and leaf water potential, ψ_l , from the 1990 experiment (dripper lines in every interrow space) are summarized in Fig. 1. Data for Day of Year (DOY) 214 and 221 are 4 and 1 d after irrigation, respectively, when the soil was wet. DOY 218 and 228 immediately preceded the next application, corresponding to drier soil conditions. Maximum transpiration rate and minimum leaf water potential were lower on the last days of the irrigation cycles. Since climatic potential transpiration computed according to Fuchs et al. (1987) during the 4 d was similar (Table 2), the change shows that soil water depletion induced stress. Average soil water potentials at a depth of 0 to 0.6 m during these days are presented in Table 2. The two drying cycles produced similar results.

For each of the 4 d of measurement, the relationship between F and ψ_l was clearly linear ($r^2 > 0.91$). Table 2 shows the parameters derived from the linear regression.

In the wet soil, plant hydraulic conductance was considerably higher than in the dry soil. The intercepts, ψ_0 , of all four regressions did not differ from each other, but

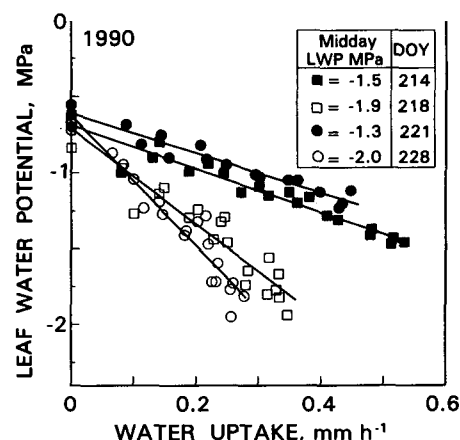


Fig. 1. Relationship between water uptake and leaf water potential of field-grown cotton plants (Na'an, 1990). Data correspond to two drying cycles. Day of year (DOY) 214 is 4 d after a water application, and DOY 221 is 1 d after irrigation. DOY 218 and DOY 228 are at the end of the drying periods. Driller lines were spread in every interrow space.

were significantly less than the bulk soil water potentials (Table 2).

In the 1991 experiment, with drippers laid in alternate interrow spaces, the (F, ψ_l) values, collected during a day did not fall along a straight line (Fig. 2). Afternoon data were below those of the morning, displaying a loop similar to that reported in the literature with transpiration data (Reicosky et al., 1982; Sharratt et al., 1983; Katerji et al., 1986). Delaying irrigation accentuated the difference between morning slopes from 1.8 (slope = $1/K_T = 1/0.55$) for data taken 1 d after irrigation (Fig. 2) to 2.6 for data 7 d later, and down to 4.0 MPa h mm⁻¹ at 9 d after an irrigation that did not replenish the water deficit in the root zone. The steepening of the slopes suggests that plant hydraulic conductance diminished during the drying cycle.

The results of the treatment with drippers in every interrow space in 1992 confirms the linear relation between ψ_l and F found in 1990. Linearity was maintained even after withholding irrigation, which dropped soil water potential to -60 kPa in the 0.3- to 0.6-m soil layer (Fig. 3, squares) with a hydraulic conductance of 0.45 mm h⁻¹ MPa⁻¹ ($r^2 = 0.81$). Linearity was also

Table 2. Linear regression parameters of the relationship between leaf water potential (ψ_l) and water uptake (F) in cotton, as presented in Fig. 1, when irrigation was applied in every interrow space.

Parameter†	Day of Year (DOY)‡			
	214	218	221	228
T_p , mm d ⁻¹	9.0	8.4	8.9	8.9
Midday ψ_l , MPa	-1.5	-1.9	-1.3	-2.0
ψ_0 , MPa	-0.05	-0.08	-0.03	-0.07
Slope, MPa h mm ⁻¹	2.88	6.28	2.62	8.56
SE of slope	0.19	0.45	0.16	0.54
ψ_0 at zero uptake, MPa	-0.69	-0.71	-0.61	-0.62
SE of ψ_0	-0.066	-0.116	-0.052	-0.110
r^2	0.925	0.912	0.939	0.929

† ψ_0 is the intercept of the ratio, T_p is the potential transpiration, and ψ_l is the soil water potential as measured at midmorning; see Eq. [1].

‡ DOY 214 is 4 d after irrigation, DOY 218 is 8 d after, DOY 221 is 1 d after, and DOY 228 is 8 d after irrigation.

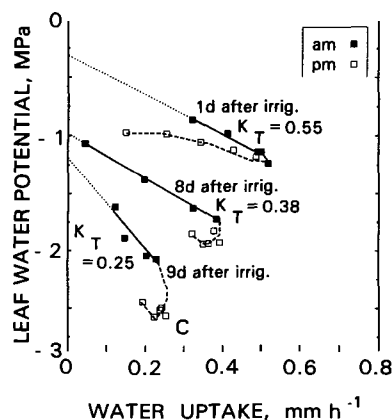


Fig. 2. Relationship between water uptake and leaf water potential of field-grown cotton plants with dripper lines laid in alternate interrow spaces (Nahal Oz, 1991). The irrigation after 9 d did not replenish the previous water depletion. K_T is the hydraulic conductance (the reciprocal of the slope) of the morning hours.

obtained 1 d after irrigation in the alternate-interrow space treatment (Fig. 3, triangles; soil water potential = -0.2 kPa, hydraulic conductance = 1.89 mm h⁻¹ MPa⁻¹, $r^2 = 0.97$). Measurements taken 7 d after irrigation in the alternate-interrow space treatment (Fig. 3, circles) developed along a clear loop, in spite of the relatively high (-30 kPa) soil water potential of the irrigated zone.

In 1993, when the soil was wet the relationship between F and ψ_l remained linear before and after noon in both treatments 1 d after irrigation (Fig. 4; soil water potential not shown). Withholding irrigation in the alternate-interrow space treatment decreased the average soil water potential (ψ_s) of the top 0.6-m soil layer to -70 kPa (Fig. 5, solid symbols). The profile of soil water potential of the wetted zone did not differ appreciably in the two treatments. Eleven days after a water application in the treatment with drippers in every interrow space (ψ_s of the top 0.6-m layer = -160 kPa; see Fig. 5), data points maintained their linear relation (Fig. 4). The slope of the line became steeper, showing that the hydraulic

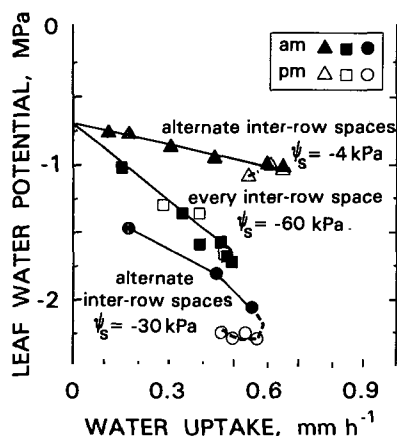


Fig. 3. Relationship between water uptake and leaf water potential of field-grown cotton plants (Nahal Oz, 1992). Squares: irrigation was withheld (soil water potential $\psi_s = -60$ kPa); triangles, 1 d after irrigation ($\psi_s = -0.2$ kPa); circles, 7 d after irrigation ($\psi_s = -30$ kPa).

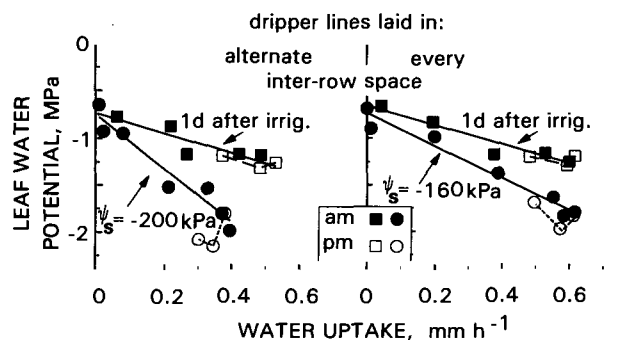


Fig. 4. The relationship between water uptake and leaf water potential of field-grown cotton plants (Nahal Oz, 1993). Solid symbols: morning hours; hollow symbols, afternoon hours. Squares: 1 d after irrigation; circles: 11 d after irrigation.

conductance decreased but the upstream potential ψ_0 remained unchanged. Similar conditions ($\psi_s = -200$ kPa in the top 0.6 m) in the treatment with drippers in alternate interrow spaces displaced afternoon data points to below those collected in the morning.

DISCUSSION

The initial idea that led to the conception of this study was to use the known relation between diurnal water uptake and leaf water potential to derive plant hydraulic conductance, K_T , and equilibrium water potential at zero uptake, ψ_0 , to characterize the effect of localized irrigation on the water status of cotton during the time interval between two irrigations. The assumption was that by using a methodology minimizing the effect of plant hydraulic capacitance, the diurnal change of soil water content in irrigated fields is sufficiently small to comply with the steady state implied by Eq. [1]. It should be noted that the assumption of steady state may not apply to all cases (see introduction, above). According to Bresler (1977), drippers in every interrow space wet more than 75% of the soil volume down to a depth of 0.6 m, while drippers in alternate interrow spaces wet only 40% of the soil to the same depth. The results show that when the water delivery design reaches a large proportion of the soil volume, the assumption of small hydraulic

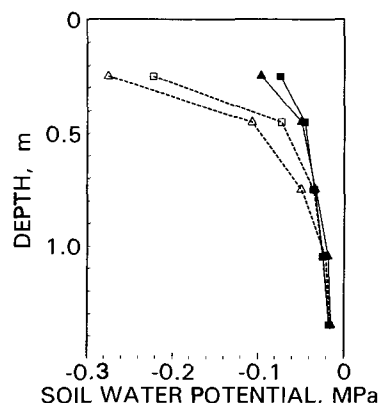


Fig. 5. Profile of soil water potential after two drying cycles (Nahal Oz, 1993). Dripper lines were spread in alternate interrow spaces (triangles) and in every interrow space (squares). Solid symbols: 6 d after irrigation; hollow symbols: 12 d after irrigation.

capacitance in plant tissues is correct and yields a linear relationship between F and ψ_1 . For the localized application of the alternate-interrow drip system, diurnal changes of water availability in the wetted volume were faster than initially thought. The link between F and ψ_1 remained apparently linear until midday, but deviated sharply from linearity in the afternoon. Since these results were obtained repeatedly over several years and different application modalities, the interpretation of the contrasting results may provide additional information on water transport through the soil-root interface.

With drippers in every interrow space, plant hydraulic conductance decreased from 0.35 and 0.38 to 0.16 and 0.12 (computed from the slopes in Table 2) when delaying irrigation allowed the soil to dry to a degree that caused water stress. Katerji et al. (1986) reached a similar conclusion by analyzing water flow in the system that includes hydraulic capacitance, but without providing a mechanistic model to explain the change. Fernandez and McCree (1991a) developed a model that incorporates capacitance and soil and plant variables to predict the interacting effects of drought on conductances. In a companion paper, Fernandez and McCree (1991b) claim that plant xylem conductance is responsible for differences between species, and that root conductance is much less limiting. Figures 1 and 5 show that soil water depletion did not affect ψ_0 , the leaf water potential at which flow in the plant ceases. A constant ψ_0 provides strong evidence that the change in conductance occurred in the water-conducting vessels of the plant.

Measurements in citrus, grown in a heavy clay soil using similar procedures as in the present experiment (Cohen et al., 1983), contradict our 1990 results that showed a variation in K_T following soil water depletion but unchanged ψ_0 . In their study, Cohen et al. (1983) found that soil water depletion lowered ψ_0 while maintaining a nearly constant value of the plant hydraulic conductance, as assumed in the linear model conductance of Lascano and Van Bavel (1984). Passioura and Munns (1984) showed that soil water depletion lowers ψ_0 , but found that hydraulic conductance remained constant in some species but decreased in others. The divergent results suggest that the hydraulic adjustment of plants may depend on the interactions between root medium and plant hydraulic characteristics. Results reported by Bristow et al. (1984) provide a clue that could explain the divergent results. They found that the lumped conductance across the soil-root interface and in the plant remained nearly constant over a wide range of soil water contents, but decreased steeply when soil water depletion fell below 25% of saturation, regardless of soil texture. The soil water potential at which the break occurred decreased as the clay content of the soil became higher.

To explain the results shown in Fig. 1, one can envision that, as the soil dries, it shrinks at the contact with roots, where water extraction is most intense. The shrinkage causes gaps interrupting the water flow through some roots in the dried segments of the soil (Herkelrath et al., 1977). In the roots with interrupted flow, the water potential was everywhere equal to ψ_1 . Where roots remained in contact with the wet soil, the entry water

potential was ψ_0 . In spite of water depletion, the entry potential would remain fairly constant, because $d\theta/d\psi_s$ (where θ is the relative soil water content) is very large in wet soils. However, because fewer conducting roots participated in the water transport, the effective hydraulic conductance of the plant appeared to have decreased. This decrease is temporary and reversible, as wetting the soil by irrigation restores the conductance to its original value. There are some indications in the literature, however, that radial root conductance may decrease when soil is dried (Yamauchi et al., 1995).

The nonlinear response in Fig. 2 and 4, corresponding to the treatment with low-density dripper distribution, appears to be a direct consequence of the small volume of wetted soil in which the root system concentrated. During a day, a sufficient number of roots lost contact with the soil to lower appreciably the effective plant hydraulic conductance, shifting (F , ψ_1) points collected during the afternoon to below those of the morning.

Since the rate of water extraction per unit volume of wetted soil is higher in this treatment, it is also possible that the water content near roots decreased to a level at which $d\theta/d\psi_s$ became small, causing a drop in ψ_0 within the day. Measurements of ψ_1 did not cover a full diurnal cycle, so we could not explore this hypothesis directly. Still, in experiments with pressurized root systems, Passioura and Munns (1984) showed that, when root media had low volumetric water content, the value of ψ_0 in the afternoon was below that of the morning.

The experimental results confirm previous findings that the values of ψ_0 are considerably lower than the matric potential of the soil (Berger, 1971; Cohen et al., 1983; Lascano and Van Bavel, 1984; Passioura and Munns, 1984; Moreshet et al., 1990; Rieger and Motisi, 1990); nonetheless, they do not provide new information regarding the diode electrical analog or the mechanical equivalent of a diode discussed by Passioura and Munns (1984). Diurnal changes of the water flow in the soil-root contact zone, as observed by Baker and Van Bavel (1988), may also explain the nonlinearity of the uptake-potential relationship. By contrast, Berger's (1971) data of ψ_0 above -0.1 MPa for plants grown in stirred solutions, and the sharp rise in ψ_0 reported by Rieger and Motisi (1990) and Hirasawa and Ishihara (1991) for plants with severed roots, suggest a biological membrane effect across the soil-root interface or within the outer layers of the conducting root tissues. It should be noted, however, that the nonlinearity may be a result of the decline in leaf water due to leaf tissue dehydration.

Whether the lower afternoon points in Fig. 2 and 5 result from a decrease of plant effective hydraulic conductance, from a change in the water potential at the entry sites across the soil-root interface, or from leaf tissue dehydration, all explanations suggest that afternoon water stress occurred in the treatment with low dripper density. Since water was applied at approximately 3-d intervals, stress periods were frequent. This stress affected total biomass production: cotton in the treatment with dripper line between every interrow space reached a height of 1.63 m, whereas height in the treatment with alternate-interrow space dripper lines was only

1.38 m. The effects on the harvested yield have not been examined.

CONCLUSIONS

The data presented in this paper show that delaying irrigation on a uniformly wetted soil decreased effective plant hydraulic conductance and caused water stress. The fact that ψ_0 remained constant supports the hypothesis of a decrease of the effective plant hydraulic conductance. This decrease was temporary and reversible, as irrigation restored conductance to its original value. Hydraulic conductance changed during the day when localized irrigation reduced the wetted soil volume from which roots extract water. It is assumed that a sufficient number of roots lost contact with the soil to lower appreciably the effective plant hydraulic conductance, shifting (F , ψ_i) points collected during the afternoon to below those of the morning. Since the rate of water extraction per unit volume of wetted soil was higher in the localized irrigation, it is also possible that the water content near roots decreased to a level at which capacitance of the soil became small, causing a drop in ψ_0 within the day. The localized irrigation caused frequent stress periods that affected biomass production. The effect on the harvested yield must still be examined.

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